

Object Detection Models

YOLOv8, Faster R-CNN, DETR

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1 Introduction

Object detection is a core problem in computer vision that focuses on identifying and locating objects within an image. It plays a vital role in many real-world applications such as autonomous driving, surveillance, medical imaging, and robotics. Over time, various model architectures have been developed to improve the speed and accuracy of detection. These models are typically grouped into three main categories:

- **One-stage detectors**, such as YOLOv8, which perform detection in a single step and are known for their real-time performance.
- **Two-stage detectors**, such as Faster R-CNN, which first propose object regions and then refine the results—typically achieving higher accuracy.
- **Transformer-based detectors**, like DETR, which introduce attention mechanisms to model global relationships in the image and simplify the detection pipeline.

This report presents a comparative analysis of these three representative models, focusing on their internal architectures, detection capabilities, and performance on benchmark datasets.

2 Datasets

2.1 COCO Dataset

(COCO) dataset is a widely adopted benchmark for assessing object detection, instance segmentation, and image captioning algorithms. In our study, we utilize the 2017 edition, which comprises:

- **Training images:** 118,287
- **Validation images:** 5,000
- **Evaluated categories:** 80 object classes

Note: While COCO annotations list 91 classes in total, only 80 of them are officially evaluated. The remaining 11 classes are included in the category index but lack labeled instances within the main dataset splits.

2.2 Pascal VOC 2007 Dataset

The **Pascal Visual Object Classes (VOC) 2007** dataset is another key benchmark commonly used to evaluate the performance of object detection models. In this project, we rely on:

- **Trainval set:** 5,011 annotated images
- **Test set:** 4,952 annotated images
- **Categories:** 20 distinct object classes

VOC 2007 offers a practical and compact dataset with varied object appearances, occlusions, and real-world imaging conditions. Its size and structure make it ideal for model prototyping and performance comparison across different architectures.

3 Architectures

3.1 YOLOV8

- **Backbone:**
 - **CBS Blocks (Conv-BN-SiLU):** Basic building block: Convolution → BatchNorm → SiLU (Swish) activation.
 - **C2f Modules:**
 - * Similar to CSP (Cross Stage Partial) modules, but more efficient.
 - * Split input, process one part with bottlenecks, then fuse back.
 - **SPPF (Spatial Pyramid Pooling – Fast):**
 - * Helps to capture multi-scale features with minimal computation.
 - * Replaces traditional SPP with a faster variant.
 - **Downsampling via Strided Convolutions or MaxPool:**
 - * Reduces spatial resolution while increasing channel depth.
- **Neck (feature extractor):** Merges feature maps from different stages of the backbone to capture information at various scales.
- **Head:** Responsible for making predictions.

3.2 Faster R-CNN

- **Backbone (Feature Extractor):**

- Typically a pre-trained CNN like ResNet-50 or ResNet-101.
- Extracts convolutional feature maps from the input image.
- Often includes Feature Pyramid Networks (FPN) to capture multi-scale features.

- **Region Proposal Network (RPN):**

- A shallow network that slides over the feature map produced by the backbone.
- For each location, it predicts:
 - * **Objectness score:** whether the region contains an object.
 - * **Bounding box offsets:** refinements to default anchor boxes.
- Outputs a set of region proposals (ROIs) with high objectness scores.

- **Anchor Boxes:**

- Predefined boxes of different scales and aspect ratios used by the RPN.
- Anchor boxes are refined using regression offsets predicted by the RPN.

- **ROI Align (Region of Interest Alignment):**

- Takes variable-sized region proposals and extracts fixed-size (e.g., 7×7) feature maps from the backbone feature map.
- ROI Align improves upon ROI Pooling by preserving spatial alignment (no rounding).

- **Head (Two Fully Connected Layers):**

- Processes the aligned ROIs with two fully connected (FC) layers.
- Outputs two branches:
 - * **Classification Branch:**
 - Uses a **Softmax** function.
 - Outputs class probabilities (including background).
 - * **Bounding Box Regression Branch:**
 - Outputs four coordinates for each class: $[t_x, t_y, t_w, t_h]$.

- These are used to refine the final bounding box locations.

- **Post-processing:**

- Applies **Non-Maximum Suppression (NMS)** to remove redundant overlapping boxes.
- Final output is a set of object detections with class labels, confidence scores, and bounding boxes.

- **Training Strategy:**

- Multi-task loss = classification loss (Softmax Cross-Entropy) + bounding box regression loss (Smooth L1).
- Trained end-to-end in a two-stage pipeline:
 1. Train RPN to generate high-quality proposals.
 2. Train classification and regression heads on the proposed regions.

3.3 DETR ([DE]tection [TR]ansformer)

- **Backbone:**

- **ResNet-50:** Used as a feature extractor. It outputs a grid of image features that are passed into the transformer.
- **Positional Encoding:** Since transformers are permutation-invariant, positional encodings (sine and cosine based) are added to the backbone features to retain spatial information.

- **Transformer Encoder-Decoder:**

- **Encoder:** A standard transformer encoder processes the flattened feature map. It consists of multi-head self-attention layers and feed-forward networks.
- **Decoder:** Uses a fixed number of learned object queries. Each query attends to the encoder output and attempts to predict an object.
- **Object Queries:** Learnable embeddings that allow the model to detect a fixed number of objects in a single forward pass.

- **Prediction Head:**

- Consists of a shared feed-forward network applied to each decoder output.

- Each query outputs a bounding box and class label.
 - DETR performs a bipartite matching (Hungarian algorithm) between predicted boxes and ground truth, enabling end-to-end optimization without anchor boxes or NMS.
- **Inference and Evaluation:**
 - Pretrained DETR (ResNet-50) from Hugging Face was used.
 - Inference was performed on the Pascal VOC 2007 test set.
 - Predictions were post-processed using the built-in DETR image processor with a confidence threshold of 0.7.
 - Results were converted to COCO format and evaluated using `pycocotools` for standard object detection metrics (AP, AR).

4 Evaluation Metrics

4.1 Intersection over Union (IoU)

Intersection over Union (IoU) quantifies the degree of overlap between a predicted bounding box and the corresponding ground truth box. It is computed as:

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}} \quad (1)$$

A higher IoU indicates better localization. Detection results are considered correct when their IoU with the ground truth exceeds a predefined threshold.

4.2 Mean Average Precision (mAP)

Mean Average Precision (mAP) is a standard metric used to evaluate the accuracy of object detection models. It combines both localization and classification performance. The mAP is computed as:

$$\text{mAP} = \frac{1}{N} \sum_{i=1}^N \text{AP}_i \quad (2)$$

Here, N is the number of object categories, and AP_i is the Average Precision for the i -th class. In our experiments, we report:

- **mAP@0.5:** Average Precision at a fixed IoU threshold of 0.5
- **mAP@0.5:0.95:** Average of AP values computed at multiple IoU thresholds from 0.5 to 0.95 in steps of 0.05

5 Results

5.1 YOLO

5.1.1 COCO Dataset

Table 1: YOLO Performance on COCO Dataset

IoU Threshold	mAP	Precision	Recall
0.5:0.95	37.2	37.4	32.0
0.5	52.1	52.6	81.1

5.1.2 Pascal VOC Dataset

Table 2: YOLO Performance on Pascal VOC Dataset

Metric	Value
mAP@0.5	57.78
mAP@0.5:0.95	45.67
Precision	59.06
Recall	62.78

5.2 Faster R-CNN

5.2.1 COCO Dataset

Table 3: Faster R-CNN Performance on COCO Dataset

IoU Threshold	mAP	Precision	Recall
0.5:0.95	34.44	37.8	42.23
0.5	53.83	46.19	52.13

5.2.2 Pascal VOC Dataset

Table 4: Faster R-CNN Performance on Pascal VOC Dataset

Metric	Value
mAP@0.5	73.35
mAP@0.5:0.95	46.387
avg_IOU	0.511
Precision	17.77
Recall	93.52

5.3 DETR

5.3.1 COCO Dataset

Table 5: DETR Performance on COCO Dataset

IoU Threshold	mAP	Precision	Recall
0.5:0.95	36.5	39.0	42.1
0.5	52.3	46.5	49.7

5.3.2 Pascal VOC Dataset

Table 6: DETR Performance on Pascal VOC Dataset

Metric	Value
mAP@0.5	55.9
mAP@0.5:0.95	41.3
avg_IOU	62.6
Precision	55.9
Recall	47.0

6 Model Comparison and Analysis

This section compares the performance of the three object detection models: YOLOv8, Faster R-CNN, and DETR—based on evaluation metrics computed on the COCO and Pascal VOC 2007 datasets.

6.1 Performance Summary

6.1.1 COCO Dataset

Table 7: Performance Comparison on COCO Dataset

Model	mAP@0.5	mAP@0.5:0.95	Precision	Recall
YOLOv8	52.1	37.2	52.6	81.1
Faster R-CNN	53.83	34.44	46.19	52.13
DETR	52.3	36.5	46.5	49.7

6.1.2 Pascal VOC Dataset

Table 8: Performance Comparison on Pascal VOC 2007 Dataset

Model	mAP@0.5	mAP@0.5:0.95	avg_IoU	Precision	Recall
YOLOv8	57.78	45.67	—	59.06	62.78
Faster R-CNN	73.35	46.39	51.1	17.77	93.52
DETR	55.9	41.3	62.6	55.9	47.0

6.2 Observations

- **YOLOv8** achieves the best recall and precision on the COCO dataset, making it ideal for real-time applications where fast and accurate detection is critical.
- **Faster R-CNN** outperforms the other models on the Pascal VOC dataset in terms of mean Average Precision (mAP@0.5), indicating strong performance in smaller, structured datasets. It also achieves the highest recall, although its precision is significantly lower, suggesting more false positives.
- **DETR** shows superior average IoU on Pascal VOC, indicating better bounding box alignment. It eliminates the need for anchor boxes and Non-Maximum Suppression (NMS), offering an end-to-end transformer-based detection pipeline.